

Reading the Current, version 2: judging Geneformer’s virtual gene edits

Lab progress report, 1 October 2026 · Kaisar Dauyey · Shinji Nakaoka

The idea. Geneformer reads each cell as a list of its genes, most unusual first, and places the cell on an internal map. We teach it to tell a patient’s tumour T cells from their normal T cells, then edit a tumour cell’s list in the computer (delete a gene, or move it to the top) and measure whether the cell moves toward that patient’s normal T cells. This is called in-silico perturbation (ISP). No real cell is touched, and the model always gives an answer.

Six checks we now apply, each learned from a real mistake (lab standard ISP-STD-1, 28 September 2026)

Check	Rule	What taught us
1. Write the test down first	Test, direction, threshold, reachable p and the outcome words, fixed and reviewed before any result	In August we read meaning into an ordering we had not planned to test
2. Check the instrument	The model must work on unseen patients; an edit that changes nothing must give zero	The July normal and tumour groups came from different studies (0 of 18 supplied both)
3. Same cells for both edits	Delete and overexpress analysed on identical cells (since 25 September)	Overexpress used 2,424 cells for every gene, delete 202 to 1,131
4. Beat look-alike genes	Each gene against at least 20 genes detected about as often; ambient-RNA genes flagged	S100A8 and S100A9 moved cells the same way under both edits in 9 of 12 comparisons
5. Beat random genes	Opposite effects of the two edits are not, on their own, evidence	Random genes do it too: rank correlation -0.593 (100 genes), -0.608 (200)
6. Count patients, use fixed words	Patients are the unit; stable in 95% of 10,000 resamplings; never “no effect”	One patient held 74.9% of the SCLC test cells

What we can say. In lung (43 patients) the model separates tumour from normal T cells in unseen patients (balanced accuracy 0.825, all 43 above chance); 13 of 34 testable T-cell genes move cells more than their look-alikes, stably; and the opposed effects of the two edits are generic. In colon (19 patients, new models) the split is learnable too (0.904, 19 of 19). **What we cannot say:** that any gene controls T-cell state; what the classifier reads (T-cell state, ambient RNA or T-cell subset mix); whether the lung models work on new lung patients.

Colon results (after review). Random genes were opposed again (rank correlation -0.25; lung -0.59) but held in only 78% of resamplings, short of the 95% required: open. Of 10 lung reference genes, 3 kept their lung direction (9 required): the lung pattern did not repeat. Tissue cannot be blamed: study, handling, models and patient numbers differ.

Still to test. Whether the opposed effects belong to the model or to lung (colon run); transfer to new lung patients (blocked: data not accessible); the same effects under an unrelated goal; cell state versus T-cell subset mix; a readable baseline from published T-cell programs (TCAT, Kotliar et al. 2025); model size (104M vs 316M parameters); positive controls and real gene-editing screens. Together these could become one methods paper, working title “Opposite by default”.

Terms. **Token:** one gene in a cell’s list. **Embedding:** a cell’s position on the model’s map. **Balanced accuracy:** how often the model is right, counting both groups equally; 0.5 is a coin toss. **Ambient RNA:** genetic material from other cells picked up during sample preparation. **Rank correlation:** from -1 (opposed) through 0 to 1. Every number is sourced in SOURCES.md.